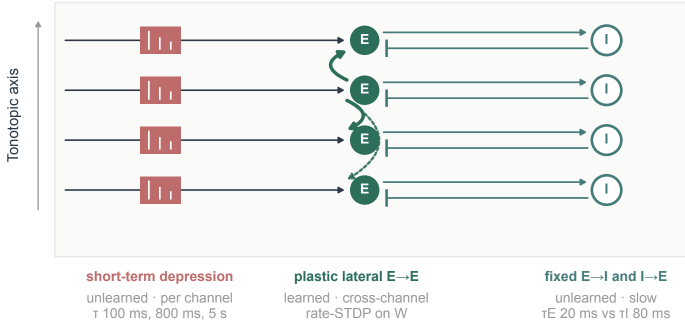
a

Layer 1 — a cortical circuit carrying three memories

one excitatory and one inhibitory cell per tonotopic channel · trained lateral connections solid, untrained dotted



State equations

$\tau_E \, dE/dt = -E + tm + W E - M_{IE} I, \quad E := \mathrm{relu}(E)$

$\tau_I \, dI/dt = -I + M_{EI} E$

$tm = A\, U\, x\, s, \quad x = \max(0, 1 - \Sigma\, w_k D_k)$

$\Delta W = \eta_{LTP} (W_{max} - W)\, E\, tr^{T} - \eta_{LTD}\, tr\, E^{T}$

M_EI and M_IE are fixed: strong on the diagonal, weak off it. Only W is plastic.

Three memories: an unlearned per-channel one, a learned cross-channel one, and a slow inhibitory state that decides how they show.

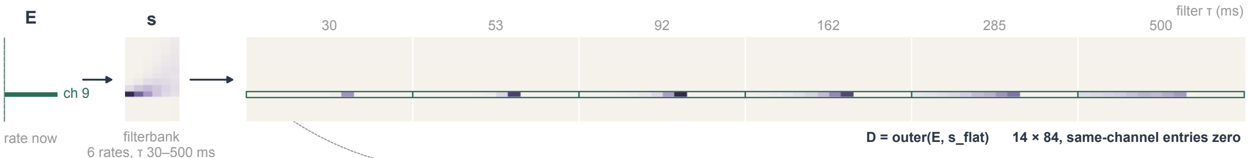
E the rate vector, one value per channel

Layer 2 reads E only — never W. That is why composition survives with layer 1 frozen or removed.

b

Layer 2 — a multiscale readout that composes relations

measured arrays, not drawings · one instant of a real Saffran stream, the final token of a four-token word · colour scaled within each map



Readout equations

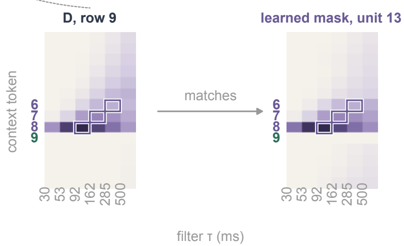
$\tau_r \, ds_r/dt = -s_r + E, \quad r = 1\ldots6$

$D = \mathrm{outer}(E, s_{flat}), \quad D_{ii} = 0$

$y = \mathrm{relu}(M \cdot D)$

$\Delta M_k = \eta\, (D/\lVert D\rVert - M_k)$, best match only

24 units, one fixed mask each.
How many commit is a result.



W3 = 6→7→8→9, held in order

token 6	τ 285 ms
token 7	τ 162 ms
token 8	τ 92 ms
token 9	firing now

Older tokens sit on strictly slower filters.
That ordering is what makes the mask a word rather than a set of tokens.